

Model Validation

Validation of Fama-French 3 Factors Model

Contents

Executive Summary	2
Purpose and Scope	2
1. Model Overview	2
2. Model Formula and Inputs.....	2
3. Model Assumptions	3
4. Tech Specification	3
4.1. Model Factors	3
4.2. Test Data Set	3
4.2.1. Data frequency	3
4.2.2. Data range	4
4.2.3 Selected Stock and Portfolio	4
4.3 Validation Tool	4
5. Testing of Pricing.....	5
5.1 Test on Period from Year 2015 to Year 2019.....	5
5.2 Sensitive Analysis – Test on Time Period with Pandemic Period	11
5.3 Sensitive Analysis – Add random noise to the FF factors	13
5.4 Sensitive Analysis – Add random noise to the GOOG price.....	14
6. Conclusion	15
6.1 Conditions for Use.....	15
6.2 Strengths and Limitations	15
7. Reference	15

Executive Summary

Validating Fama-French 3 Factors Model to practice model validation techniques. To verify that models performing as expected, in line with design objectives and business uses. Ensures models sound, identifies potential limitations in assumptions, and assesses their possible impact. addresses risks within a quantitative model.

Purpose and Scope

The purpose of this study is to explore the issues surrounding the validation of pricing and risk models in financial institutions. Testing the classic assets pricing model, Fama-French Three-factor Model and try to find out the strengths, limitations, weaknesses and provide suggestion of conditions for use.

1. Model Overview

The Fama-French Three-factor Model is an extension of the Capital Asset Pricing Model (CAPM). The Fama-French model aims to describe stock returns through three factors:

- (1) market risk,
- (2) the outperformance of small-cap companies relative to large-cap companies,
- (3) the outperformance of high book-to-market value companies versus low book-to-market value companies.



The rationale behind the model is that high value and small-cap companies tend to regularly outperform the overall market.

2. Model Formula and Inputs

The Fama/French factors are constructed using the 6 value-weight portfolios formed on size and book-to-market. The mathematical representation of the Fama-French three-factor model is:

$$r = r_f + \beta_1(r_m - r_f) + \beta_2(SMB) + \beta_3(HML) + \varepsilon$$

Where:

- r = Expected rate of return
- r_f = Risk-free rate
- β = Factor's coefficient (sensitivity)

- $(r_m - r_f)$ = Market risk premium
- SMB (Small Minus Big) = Historic excess returns of small-cap companies over large-cap companies
- HML (High Minus Low) = Historic excess returns of value stocks (high book-to-price ratio) over growth stocks (low book-to-price ratio)
- ϵ = Risk

SMB (Small Minus Big) is the average return on the three small portfolios minus the average return on the three big portfolios,

$$SMB = 1/3 (\text{Small Value} + \text{Small Neutral} + \text{Small Growth}) - 1/3 (\text{Big Value} + \text{Big Neutral} + \text{Big Growth}).$$

HML (High Minus Low) is the average return on the two value portfolios minus the average return on the two growth portfolios,

$$HML = 1/2 (\text{Small Value} + \text{Big Value}) - 1/2 (\text{Small Growth} + \text{Big Growth}).$$

$R_m - R_t$, the excess return on the market, value-weight return of all CRSP firms incorporated in the US and listed on the NYSE, AMEX, or NASDAQ that have a CRSP share code of 10 or 11 at the beginning of month t , good shares and price data at the beginning of t , and good return data for t minus the one-month Treasury bill rate (from Ibbotson Associates).

3. Model Assumptions

The three-factor model is designed based on the assumption that riskier investments require higher returns. According to the Fama-French three-factor model, over the long-term, small companies outperform large companies, and value companies beat growth companies.

The studies conducted by Fama and French revealed that the model could explain more than 90% of diversified portfolios' returns.

4. Tech Specification

4.1. Model Factors

Using Fama/French 3 Research Factors as the model factor values. And the factors value used in the study downloaded from official website of data library of Ken French.

4.2. Test Data Set

4.2.1. Data frequency

Using monthly pricing data to test the model.

As it avoids introducing too much microstructure market noise, such as the mismatch between the bid-ask spread and the unbalanced order flow. These are not caused by the behavior of investors, but caused by friction in the economic market. In addition, for better quality of validation, the market need enough time to digest all the information.

4.2.2. Data range

Conduct the test by pricing data of selected stock and portfolio at range of recent 5 years, from January 2014 to December 2019.

Plus, as a comparison, using the data of first 3 months of year 2020, which is the pandemic time period, to test model performance under extreme situation.

4.2.3 Selected Stock and Portfolio

Alphabet Inc. (GOOG)

Facebook, Inc. (FB)

Amazon.com, Inc. (AMZN)

Tesla, Inc. (TSLA)

SPDR Portfolio S&P 500 Growth ETF

Invesco S&P 500 High Dividend Low Volatility ETF (SPHD)

4.3 Validation Tool

Excel Data Analysis, Regression.

5. Testing of Pricing

5.1 Test on Period from Year 2015 to Year 2019.

1

Profile
 Data Frequency Monthly
 Data Range 5 Years
 Start Date 1/1/2015
 End Date 12/1/2019
 Data Source Alphabet Inc. (GOOG)

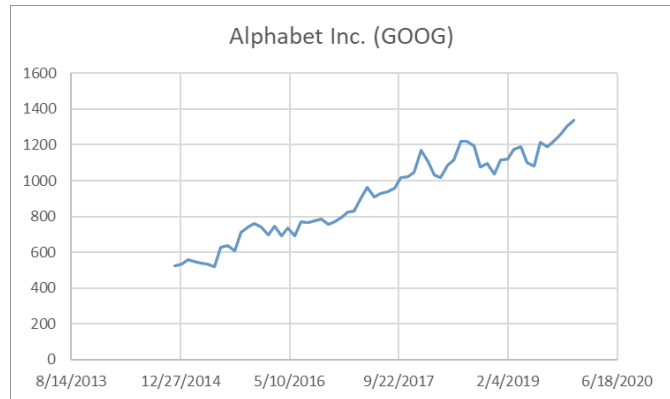
SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.665201833
R Square	0.442493479
Adjusted R Square	0.412627058
Standard Error	0.044706137
Observations	60

ANOVA

	df	SS	MS	F	Significance F
Regression	3	0.088834006	0.029611335	14.81575185	3.22763E-07
Residual	56	0.111923768	0.001998639		
Total	59	0.200757774			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.004606508	0.006043218	0.762260759	0.449103158	-0.007499512	0.016712527	-0.007499512	0.016712527
Mkt-RF	1.038737611	0.169247406	6.13739162	9.0775E-08	0.699694316	1.377780907	0.699694316	1.377780907
SMB	-0.701800057	0.256101729	-2.740317531	0.008220476	-1.214833468	-0.188766646	-1.214833468	-0.188766646
HML	-0.457645683	0.224083839	-2.042296692	0.045841511	-0.906539553	-0.008751813	-0.906539553	-0.008751813



Result Interpretation:

The value of R square shows that the model overall performs well. From the p-values, we can tell that all of these three fama factors are significant on the 95% confidence level. Note that the linear coefficient of excess market return(Mkt – RF) is roughly 1, which means the beta is close to 1. On the other hand, the linear coefficient of SMB and HML are negative, which makes sense. Because GOOG has a huge market capitalization(about 900B according to Yahoo Finance), and its B/P ratio is low. Hence GOOG's return's main contributor is, apparently, the market excess return factor. Besides, we see that the intercept is not significant, which means our model fits the historical data well.

2

Profile
 Data Frequency Monthly
 Data Range 5 Years
 Start Date 1/1/2015
 End Date 12/1/2019
 Data Source Facebook, Inc. (FB)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.550694635
R Square	0.303264581
Adjusted R Square	0.26593947
Standard Error	0.060565113
Observations	60

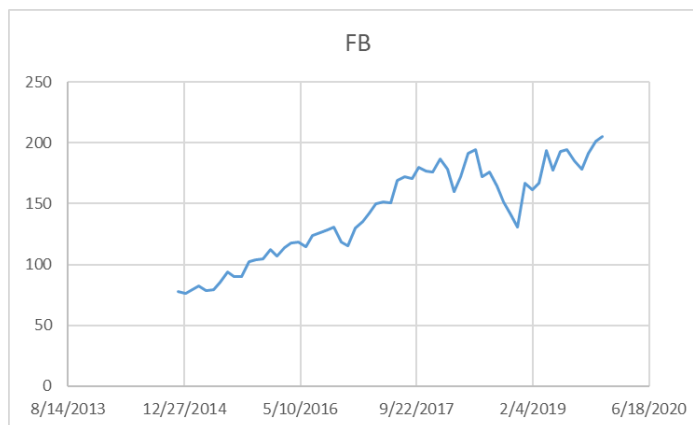
ANOVA

	df	SS	MS	F	Significance F
Regression	3	0.089410165	0.029803388	8.124947726	0.000139602
Residual	56	0.205415445	0.003668133		
Total	59	0.29482561			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.006540138	0.008186978	0.798846476	0.427754587	-0.009860349	0.022940626	-0.009860349	0.022940626
Mkt-RF	1.024107994	0.229285931	4.466510396	3.90606E-05	0.564793081	1.483422908	0.564793081	1.483422908
SMB	-0.116731763	0.346950802	-0.336450477	0.737788738	-0.811757737	0.578294211	-0.811757737	0.578294211
HML	-0.540965134	0.303574943	-1.781982165	0.08017419	-1.149098821	0.067168553	-1.149098821	0.067168553

Result Interpretation:

The value of R square indicates that the model's performance is great overall. The regression results here is quite similar to that of GOOG. From p-values, we can see that only excess market return is significant. It is the largest contributor to its historical return. As Facebook is also a company with huge market capitalization and low BP ratio, it is understandable that SMB and HML's linear coefficients are non-significant.



3.

Profile

Data Frequency Monthly
Data Range 5 Years
Start Date 1/1/2015
End Date 12/1/2019
Data Source Amazon.com, Inc. (AMZN)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.712710181
R Square	0.507955802
Adjusted R Square	0.481596291
Standard Error	0.059238101
Observations	60

ANOVA

	df	SS	MS	F	Significance F
Regression	3	0.20286732	0.06762244	19.2703047	1.04206E-08
Residual	56	0.196512546	0.003509153		
Total	59	0.399379866			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.014745843	0.008007597	1.841481709	0.0708495	-0.001295301	0.030786988	-0.001295301	0.030786988
Mkt-RF	1.535507873	0.22426216	6.8469325	6.19411E-09	1.086256782	1.984758964	1.086256782	1.984758964
SMB	-0.644231869	0.339348934	-1.898434925	0.062796441	-1.324029472	0.035565734	-1.324029472	0.035565734
HML	-0.901145093	0.296923462	-3.034940676	0.003645344	-1.495954263	-0.306335923	-1.495954263	-0.306335923



Result Interpretation:

The value of R square indicating the model overall performs well.

And p-value indicating all 3 factors are significant to the model.

The factor, Mkt-RF, from the t-statics, indicating it is the most influential factor. Also, as the coefficient of it is a positive number, and approximately equal to 1.54, which indicating it's stock price growing more than 50% faster than the universal market.

4.

Profile
 Data Frequency Monthly
 Data Range 5 Years
 Start Date 1/1/2015
 End Date 12/1/2019
 Data Source Tesla, Inc. (TSLA)

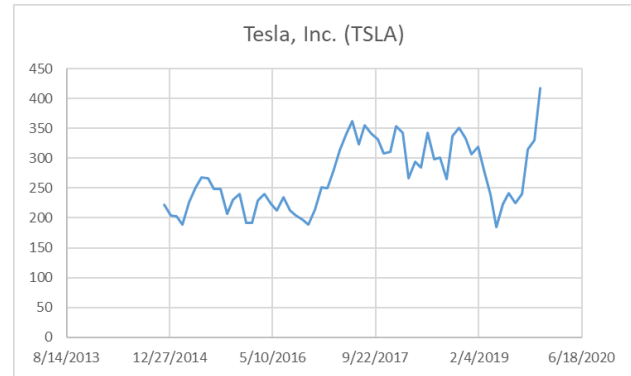
SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.188555816
R Square	0.035553296
Adjusted R Square	-0.016113492
Standard Error	0.123500089
Observations	60

ANOVA

	df	SS	MS	F	Significance F
Regression	3	0.031486486	0.010495495	0.688126691	0.563088366
Residual	56	0.854127225	0.015252272		
Total	59	0.885613711			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.011232743	0.016694305	0.67284877	0.503810706	-0.022209969	0.044675454	-0.022209969	0.044675454
Mkt-RF	0.664200145	0.467543628	1.420616398	0.160972548	-0.272402288	1.600802578	-0.272402288	1.600802578
SMB	-0.186045651	0.707477497	-0.262970414	0.793538499	-1.603293381	1.23120208	-1.603293381	1.23120208
HML	0.113594893	0.619028518	0.183505104	0.855064536	-1.126468241	1.353658026	-1.126468241	1.353658026



Result Interpretation:

The value of R square indicating the model overall performs bad. And p-value indicating all 3 factors are not significant to the model.

To further investigate the reason why the model performed bad on explain the Tesla stock price, we looked at the price trend graph. From the graph, the trend of Tesla does have the same trend with the overall market. So that we can explain why the factor, Mkt-RF, does not significant.

Moreover, the other two factors SMB and HML does not significant to the model, the model only explains very few variances of the data. Concluded, the model failed to predict the Tesla price, due to the Tesla stock price behavior doesn't seem to relate to the whole market situation, it might rely on their own company performance.

5.

Profile

Data Frequency Monthly
 Data Range 5 Years
 Start Date 1/1/2015
 End Date 12/1/2019
 Data Source SPDR Portfolio S&P 500 Growth ETF
 Category Large Growth ALL SIGNIFICANT

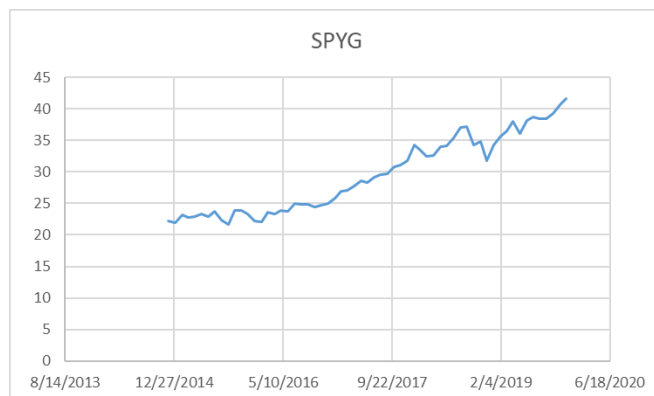
SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.978072014
R Square	0.956624864
Adjusted R Square	0.954301196
Standard Error	0.007774263
Observations	60

ANOVA

	df	SS	MS	F	Significance F
Regression	3	0.074646123	0.024882041	411.6874149	4.12321E-38
Residual	56	0.003384593	6.04392E-05		
Total	59	0.078030716			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.000130177	0.001050897	0.123872062	0.901860126	-0.001975023	0.002235377	-0.001975023	0.002235377
Mkt-RF	0.995957442	0.029431614	33.8397154	5.71072E-39	0.936998834	1.054916049	0.936998834	1.054916049
SMB	-0.210325477	0.044535319	-4.722666909	1.60264E-05	-0.299540442	-0.121110513	-0.299540442	-0.121110513
HML	-0.277756913	0.038967504	-7.127911233	2.12715E-09	-0.355818204	-0.199695621	-0.355818204	-0.199695621



Result Interpretation:

The value of R square indicating the model overall performs extremely well, more than 95% variance been covered by the model. And p-value indicating all 3 factors are significant to the model. Mkt-RF contribute extraordinarily to the model, most of the variance of the portfolio price change been captured by the factor Mkt-RF.

Hence, while investing SPYG, we are confident to use the model as one of the decision-making elements in the trading strategy.

6.

Profile

Data Frequency Monthly
 Data Range 5 Years
 Start Date 1/1/2015
 End Date 12/1/2019
 Data Source Invesco S&P 500 High Dividend Low Volatility ETF (SPHD)
 Category Large Value

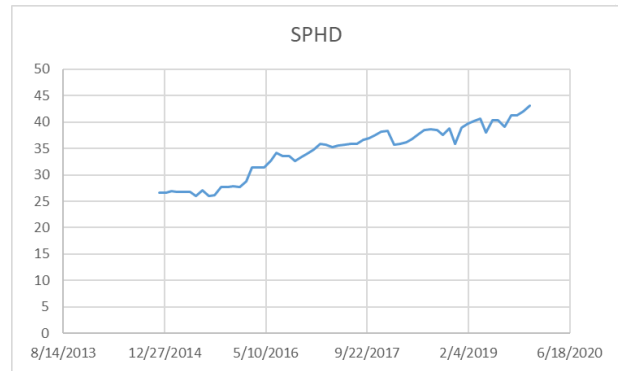
SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.784525174
R Square	0.615479748
Adjusted R Square	0.594880449
Standard Error	0.019615339
Observations	60

ANOVA

	df	SS	MS	F	Significance F
Regression	3	0.034488492	0.011496164	29.8786741	1.14481E-11
Residual	56	0.021546645	0.000384762		
Total	59	0.056035138			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.002149612	0.002651532	0.810705555	0.420966882	-0.003162045	0.007461269	-0.003162045	0.007461269
Mkt-RF	0.674383915	0.074259272	9.081477539	1.33201E-12	0.525624717	0.823143113	0.525624717	0.823143113
SMB	-0.036097836	0.112367619	-0.321247668	0.749218639	-0.261197225	0.189001554	-0.261197225	0.189001554
HML	0.119770495	0.098319397	1.218177689	0.228264037	-0.077186923	0.316727914	-0.077186923	0.316727914



Result Interpretation:

The value of R square indicating the model overall performs well. And from p-value, indicating the factor Mkt-RF is significant.

So that, the regress results the only one factor, Mkt-RF, is good enough to predict price of SPHD.

Hence, while investing SPHD, the large size and value type of portfolio, the factor Mkt-RF itself is capable of describe the price behavior.

5. 2 Sensitive Analysis – Test on Time Period with Pandemic Period

Explore if introduce data point with big volatility could affect the model performance.

1.

Data Frequency Monthly
 Data Range 5 Years, included special time period
 Start Date 1/1/2015
 End Date 3/1/2020
 Data Source Alphabet Inc. (GOOG)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.699937781
R Square	0.489912898
Adjusted R Square	0.463976265
Standard Error	0.044848149
Observations	63

ANOVA

	df	SS	MS	F	Significance F
Regression	3	0.113976572	0.037992191	18.88883997	1.04864E-08
Residual	59	0.118670032	0.002011356		
Total	62	0.232646604			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.004043837	0.005935053	0.681348064	0.498317019	-0.007832176	0.01591985	-0.007832176	0.01591985
Mkt-RF	1.132815992	0.151065907	7.498819643	3.86349E-10	0.83053381	1.435098174	0.83053381	1.435098174
SMB	-0.68869001	0.250807337	-2.745892594	0.007987185	-1.190554332	-0.186825687	-1.190554332	-0.186825687
HML	-0.299545209	0.188485194	-1.589224076	0.117354695	-0.676703211	0.077612794	-0.676703211	0.077612794



Result Interpretation:

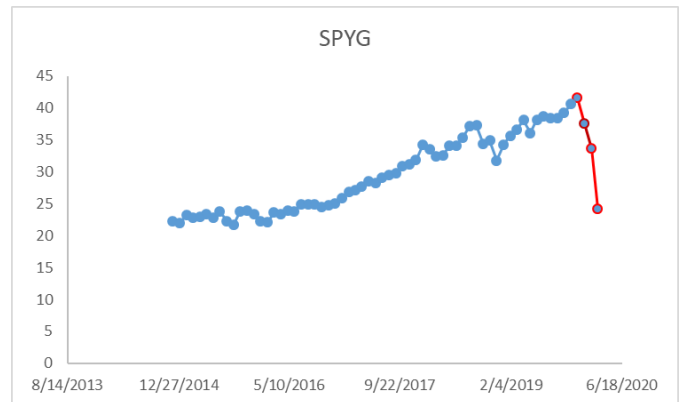
As the model works very well with the GOOG data in previous analysis, we introduce 3 more abnormal observations in the test dataset, to evaluate how much performance of the model will be affected. Add the price history in pandemic period, we can see the regress result, R square increased from 44% to 49%. The model perfectly able to transfer the market information to predict the stock price of google. It works stable to Google stock.

2.

Profile
 Data Frequency Monthly
 Data Range 5 Years, included special time period
 Start Date 1/1/2015
 End Date 3/1/2020
 Data Source SPDR Portfolio S&P 500 Growth ETF
 Category Large Growth

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.902802635
R Square	0.815052598
Adjusted R Square	0.805648493
Standard Error	0.024030625
Observations	63



ANOVA

	df	SS	MS	F	Significance F
Regression	3	0.150148003	0.050049334	86.66987271	1.3426E-21
Residual	59	0.034070786	0.000577471		
Total	62	0.18421879			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-0.002780692	0.003180132	-0.874395173	0.385448564	-0.009144121	0.003582737	-0.009144121	0.003582737
Mkt-RF	1.177787797	0.080944438	14.55057106	3.52303E-21	1.015818351	1.339757244	1.015818351	1.339757244
SMB	-0.117934589	0.134388092	-0.877567256	0.38373821	-0.386844539	0.150975361	-0.386844539	0.150975361
HML	0.183908172	0.100994516	1.820971868	0.073682916	-0.018181387	0.385997731	-0.018181387	0.385997731

Result Interpretation:

As the model works very well with the SPYG data in previous analysis, we introduce 3 more abnormal observations in the test dataset, to evaluate how much performance of the model will be affected.

The value of R square indicting the model overall performs extremely well, even of it drops to 81% from 95%. And p-value indicating only Mkt-RF factors are significant to the model. The other two factors not contributing anymore.

Hence, again while investing SPYG, we are confident to use the model as one of the decision-making elements in the trading strategy.

5.3 Sensitive Analysis – Add random noise to the FF factors

Profile
Data Frequency Monthly
Data Range 5 Years
Start Date 1/1/2015
End Date 12/1/2019
Data Source Alphabet Inc. (GOOG)
Add with 10% noise to the FF Factors

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.651098918
R Square	0.4239298
Adjusted R Square	0.393068897
Standard Error	0.045446115
Observations	60

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	3	0.085113823	0.028371274	13.7367916	7.91747E-07
Residual	56	0.115659567	0.002065349		
Total	59	0.200773391			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	0.004265127	0.006171207	0.691133353	0.492336925	-0.008097286	0.01662754	-0.008097286	0.01662754
Mkt-RF	1.033379994	0.174630213	5.917532692	2.06952E-07	0.683553641	1.383206348	0.683553641	1.383206348
SMB	-0.69789839	0.266239397	-2.621318997	0.011256964	-1.231239991	-0.164556788	-1.231239991	-0.164556788
HML	-0.462689117	0.228984126	-2.020616564	0.048111389	-0.921399441	-0.003978793	-0.921399441	-0.003978793

The model performs well even with the noised factors.

5.4 Sensitive Analysis – Add random noise to the GOOG price

Profile
 Data Frequency Monthly
 Data Range 5 Years
 Start Date 1/1/2015
 End Date 12/1/2019
 Data Source Alphabet Inc. (GOOG)
 Add with 10% noise to the price

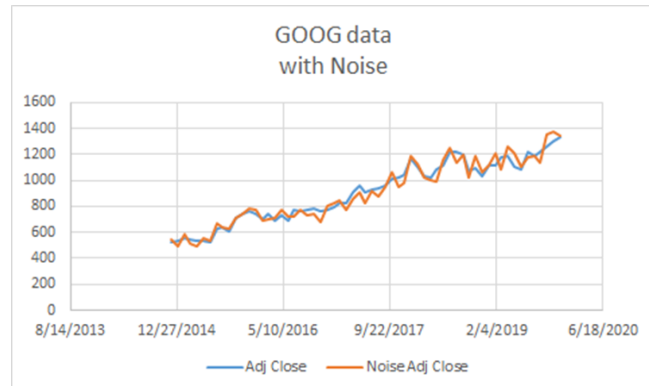
SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.665174988
R Square	0.442457764
Adjusted R Square	0.41258943
Standard Error	0.044704384
Observations	60

ANOVA

	df	SS	MS	F	Significance F
Regression	3	0.088814177	0.029604726	14.81360704	3.23331E-07
Residual	56	0.111914987	0.001998482		
Total	59	0.200729164			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.004562132	0.00604298	0.754947358	0.453444001	-0.007543412	0.016667677	-0.007543412	0.016667677
Mkt-RF	1.038114881	0.169240766	6.133952846	9.19561E-08	0.699084887	1.377144876	0.699084887	1.377144876
SMB	-0.702003217	0.256091682	-2.741218348	0.008200677	-1.215016501	-0.188989932	-1.215016501	-0.188989932
HML	-0.459076572	0.224075048	-2.048762571	0.04518273	-0.907952832	-0.010200311	-0.907952832	-0.010200311



The model performs well even with the noised price.

6. Conclusion

6.1 Conditions for Use

We could apply the Fama model to any stock or any portfolio, as long as we have access to enough data to compute the three factors. The factors could be different depending on which market our portfolio is in. Select correct market index and suitable universe is critical.

6.2 Strengths and Limitations

Overall, the model can fit the stock returns well. For the market portfolios, such as many etf based on S&P 500, as expected, the excess market return is significant and usually is the largest contributor to the portfolio historical performance. The model fits amazingly. Furthermore, the model also explain pretty well the historical performance of companies of super huge market capitalization such as GOOG and FB.

Yet we cannot deny that this model only includes three factors. Hence, it cannot fit some companies perfectly if their performance depends much on their own idiosyncratic return, such as TSLA.

Due to the limit of time, we did not test the model on a wider range of stocks. If possible, we think testing it on more small market capitalization stocks will be beneficial.

7. Reference

https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

<https://finance.yahoo.com/>